

REPORT FOR OPEN LAB PROJECT
HUMAN BODY LIFT AND DRAG AT DIFFERENT HIP BEND ANGLES

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Submitted to: Prof. Jun Chen

1 INTRODUCTION

Understanding lift and drag characteristics of a human body is important for ensuring safe travel through headwinds. When drag force on the human body becomes unreasonably large, the person is likely to be blown backward or fall. As lift force also increases, the friction of the person's feet against the ground decreases dramatically until they completely lose control and get blown away by the wind. Our team chose to investigate these effects because they are invisible forces that directly impact the lives of people walking outside and may raise safety concerns with the elderly, young, or lightweight. We plan to quantify the lift and drag effects on a small-scale model at varying headwind speeds and hip bend angles in the POTR160 Wind Tunnel.

To understand this project, a fundamental understanding of lift and drag phenomena on a bluff body, partial similarity, and nondimensional numbers must be commanded. Lift and drag are two aerodynamic phenomena that govern the impact of fluid flow on an object's momentum. As a result of ram air (i.e., air striking a surface), a body that is not symmetric about the impact axis will experience both a lift force pulling the body vertically and a drag force pushing the body horizontally. On a bluff body with a short length parallel to the flow (e.g., a standing human), most of the drag takes the form of form drag, which is governed by 1D momentum conservation.

However, experimentation on a real human or a full-scale human model presents difficulties that are beyond the team's abilities to work around. Therefore, it is necessary to leverage incomplete similarity to experiment on approximate small-scale models. However, because shrinking model size requires an inverse increase in experimental wind speed, it becomes unrealistic to match the experimental Reynolds number with the Reynolds number of the theoretical full-scale counterpart, invalidating the dynamic similarity condition of complete similarity. To validate the experiment, partial similarity must be applied by running tests up to a wind speed where the important nondimensional numbers (e.g., lift and drag coefficients) change minimally with further increase in wind speed. This phenomenon indicates that the flow is fully turbulent and that tests with higher wind speeds and Reynolds numbers will yield similar results. Partial similarity is a necessary tool to ensure the feasibility of the experiment.

Necessary assumptions associated with this experiment include Reynolds number independence past the largest-tested Reynolds number, quasi-static model, similar geometry and deformation, negligible ceiling and wall effects, negligible distance between the experimental models' feet and the ground plate, and air incompressibility. Reynolds number independence is necessary for the extrapolation of the lift and drag coefficients in partial similarity analysis. The theoretical human model must also be quasi-static with negligible deformation for the small-scale experimental model to meet geometric and kinematic similarity requirements. Ceiling and wall effects of the wind tunnel must also be neglected to allow for kinematic similarity with the streamlines near the wind tunnel's ceiling and walls. Distance between the model's feet and the ground plate must be neglected to support kinematic similarity requirements near the ground. Air incompressibility also must be assumed to neglect the effects of variable density and shockwaves.

2 OBJECTIVE

The objectives of this project are to (1) measure lift and drag coefficients of three small-scale human body models at $f = 30\text{--}60$ Hz ($U \approx 22.67\text{--}34.01$ m/s) and hip bend angle $\theta \in \{150^\circ, 165^\circ, 180^\circ\}$ using a dynamometer, (2) identify which hip bend angle produces the least lift and the least drag for each wind speed, and (3) identify the minimum wind speed where either

liftoff occurs (full scale lift force ($F_{l,fs}$) $\geq$ average human weight (W_{fs})) or the person slips backward because of the wind (full scale drag force ($F_{d,fs}$) $\geq$ average human tractive limit ($F_{tractivelimit}$)) at all hip bend angles.

3 METHOD

3.1 Experiment Facility

The experiment is performed in a wind tunnel located at the ME30801 lab in A. A. Potter Engineering Center, which is a low-speed open-circuit wind tunnel (ELD 402B, **Fig. 1**). The dimensions of its test section are 30.5 cm wide by 30.5 cm high by 61 cm long. It can be operated at a range of wind speeds from 3.0 to 48.7 m/s. A centrifugal, belt-driven fan with a 12.40-inch wheel diameter is driven by a 10 HP motor to move the air. A variable frequency drive (VFD) is used to adjust the rotating of the fan at different speeds and thus adjusts the wind speed in the test section.

Given the approximate mounting height on the dynamometer of 15 cm with about 3.37 cm of clearance at the bottom for a height-adjustable table, the experiment will be able to be conducted with at least 9.67 cm of head clearance, which is expected to be enough to avoid significant ceiling effects while encouraging ground effects.

Additionally, because the team is using incomplete similarity with the bluff body model, the wind tunnel's speed range is sufficient for the execution of this experiment.

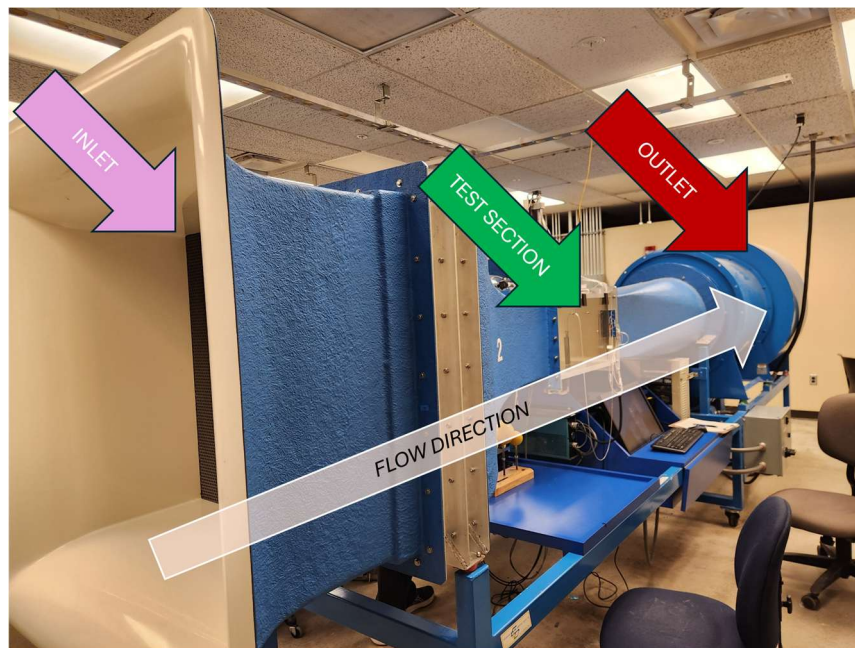


Fig. 1 Potter Engineering Center ME30801 Lab wind tunnel diagram.

3.2 Test Model

Fig. 9 shows the CAD drawings for each experimental model, while **Fig. 10** shows the models assembled into the wind tunnel. The models are 3D printed out of PLA. They are then mounted in the wind tunnel by slipping the oversized hole over the 3/8" threaded rod on the dynamometer,

then lodging it in place by nuts on either side. Several pieces of tape are wrapped over the nut to mimic clothing while reducing the aerodynamic effects associated with the nut.

Models are designed to have approximate dimensions of 8.83 cm torso length and 8.63 cm hip height, serving as a 1:10 scale model of an average human with a torso length of 88.295 cm and hip height of 86.285 cm (Gordon, 2006). These dimensions and the bend angles result in characteristic lengths of 0.1743 m for the 180° model, 0.1710 m for the 165° model, and 0.1625 m for the 150° model, where the characteristic length is taken as the height of the model normal to the floor.

3.3 Instrumentation

Tab. 1 lists the instrumentation required for the experiment. A dynamometer is used to measure the drag force experienced by the attached test model. It uses a linear variable differential transformer (LVDT) to convert linear deflection at the bottom of the apparatus structure to a high-resolution voltage. The measured voltage V_{total} can then be related to a drag force F_d between 0 and 100 N through Equation 1, where $V_{0,offset,total}$ is the voltage reading at zero wind speed with the full setup, V_{ts} is the voltage reading at test condition without the model mounted on the test stand, $V_{0,offset,ts}$ is the voltage reading at zero wind speed without the model mounted on the test stand, and K_d is the drag calibration constant. The same can be done to find the lift force F_l using Equation 2.

Tab. 1 Required instrumentation for experiment.

Instrument	Model/Type	Measured	Range	Accuracy Resolution	Available (Yes/No + Source)
LVDT Dynamometer	Unpublished	Lift & Drag	0–100 N	0.01 N	Yes (ME308 Lab)
Mercury Thermometer	Unpublished	Ambient Temperature	Unpublished	Analog	Yes (ME308 Lab)
Mercury Barometer	Unpublished	Ambient Pressure	680–780 mmHg	Analog	Yes (ME308 Lab)

$$F_d = K_d \left((V_{total} - V_{0,offset,total}) - (V_{ts} - V_{0,offset,ts}) \right) \quad (1)$$

$$F_l = K_l \left((V_{total} - V_{0,offset,total}) - (V_{ts} - V_{0,offset,ts}) \right) \quad (2)$$

The ideal gas law is used to determine the density ρ from atmospheric pressure P_{atm} , molecular mass of air M , universal gas constant R , and temperature T (Equation 3). Therefore, air in the wind tunnel must be assumed to be ideal.

$$\rho = \frac{P_{atm}M}{RT} \quad (3)$$

Temperature is measured using a mercury thermometer, and atmospheric pressure is measured using a mercury barometer and corrected by a provided table.

3.4 Experimental Conditions

The targeted free stream velocities range from 22.67–34.01 m/s and correspond to frequencies of 30–60 Hz. **Tab. 3** lists the experimental VFD frequencies, free stream velocities, and Reynolds numbers used in the experiment while **Tab. 2** shows the ambient air conditions on each of the test days. Molar mass and dynamic viscosity of air were both found from Engineering Toolbox. Equation 4 is used to find the Reynolds number of each experimental setup.

$$Re = \frac{\rho U L_c}{\mu} \quad (4)$$

Tab. 2 Test day conditions.

Parameter	Test Day 1	Test Day 2
Test Models (Hip Bend Angles)	180°, 165°	150°
Temperature (T , °C)	21.0	19.9
Uncorrected Pressure (mmHg)	751.7	744.0
Corrected Pressure (P , mmHg)	754.3	746.5
Molar Mass of Air (M , kg/kmol)	28.9647	28.9647
Air Density (ρ , kg/m ³)	1.1909	1.1787
Dynamic Viscosity (μ , Pa·s)	$18.17 \cdot 10^{-6}$	$18.12 \cdot 10^{-6}$

Tab. 3 Experimental conditions.

Model	Test Day	VFD Frequency (f , Hz)	Free Stream Velocity (U , m/s)	Characteristic Length (L_c , m)	Reynolds Number (Re)
Test Stand Only	1	0, 30–60	0, 22.67–34.01	0	0
180° Hip Bend	1	0, 30–60	0, 22.67–34.01	0.1743	0, 258981.8180–388529.8469
165° Hip Bend	1	0, 30–60	0, 22.67–34.01	0.1710	0, 254078.5478–381173.8602
Test Stand Only	2	0, 30–60	0, 22.67–34.01	0	0

150° Hip Bend	2	0, 30–60	0, 22.67–34.01	0.1625	0, 239634.8489–359505.1262
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3.5 Procedures

First, measure and record the atmospheric pressure using the barometer and the temperature using the thermometer, then ensure the wind tunnel test section, intake, and exit are clear. Turn on the LVDT, open the ME30801 folder, and launch the MATLAB “Force Measurement Dynamometer” app. Record the drag and lift scaling factors, initialize the program, and save the data file. With the wind tunnel off, collect baseline data at 0 Hz, then turn on the tunnel and incrementally increase the frequency from 30 Hz to 60 Hz in 5 Hz steps, allowing the system to stabilize for 10 seconds before recording data at each step. After completing the test stand-only measurements, turn off the tunnel and repeat the procedure for each test model (180°, 165°, and 150° bends), ensuring proper positioning before each run and saving each dataset with appropriate filenames. Finally, email all collected data files to teammates.

4 RESULTS

To determine lift and drag coefficients, lift and drag scaling factors (K_l, K_d), lift and drag force voltage data for each test (V_{total}), and test stand drag voltage were read from the experimental data. Then, using Equations 1 and 2, lift and drag force were calculated. Each respective force was used to calculate their coefficient through Equation 5, with C representing the Coefficient, F representing the force, ρ representing the density of the air, v representing the free stream velocity of the air, and A representing the planform area of the model.

$$C = \frac{2F}{\rho v^2 A} \quad (5)$$

Graphing the drag and lift coefficients against the Reynolds numbers at each point yields **Fig. 2** and **Fig. 3**. Using the standard deviation from the 10 measurements, as well as the 0.01 N uncertainty of the LVDT, the 95% confidence interval error bars were calculated.

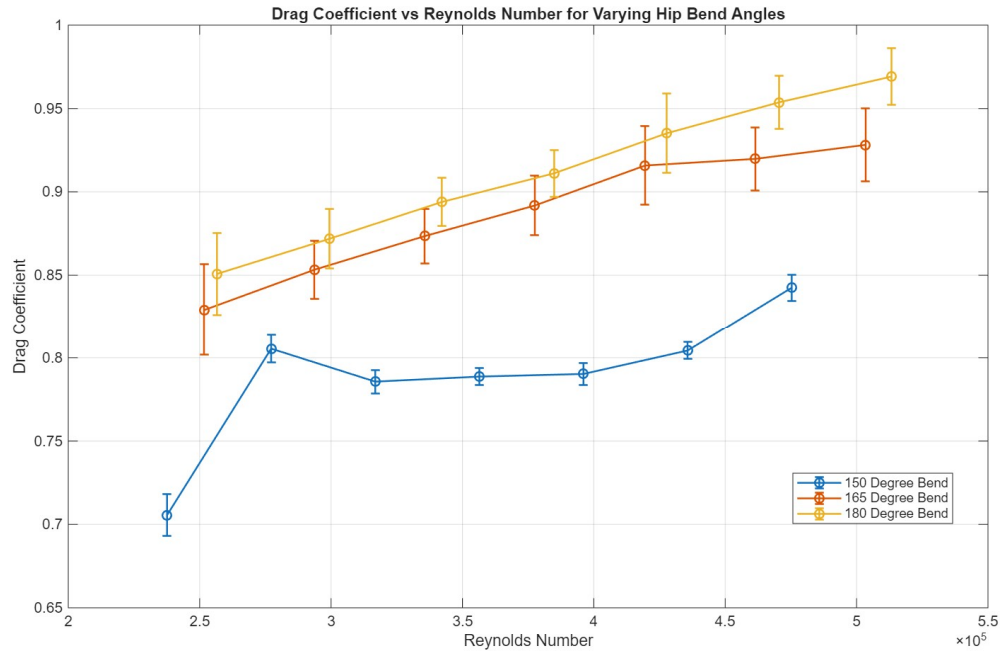


Fig. 2 Drag coefficient versus Reynolds number for varying hip bend angles. Error bars indicate the 95% confidence interval (CI) for the drag coefficient.

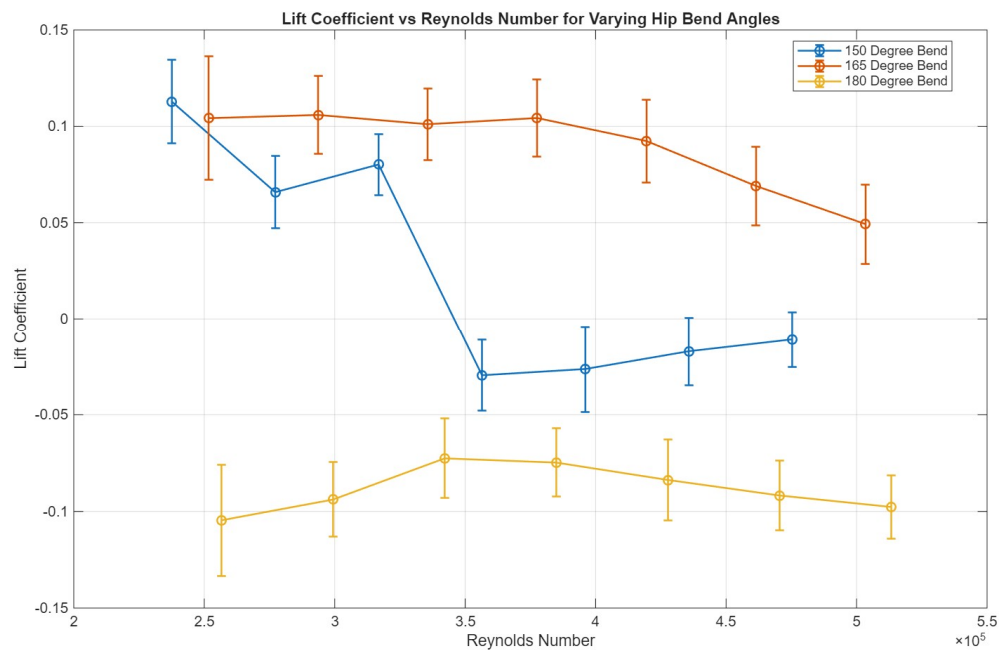


Fig. 3 Lift coefficient versus Reynolds number for varying hip bend angles. Error bars indicate the 95% confidence interval (CI) for the lift coefficient.

As expected, the 180° hip bend had the highest drag coefficients as well as the lowest lift coefficients. In fact, the lift coefficient was negative, meaning that the wind force was pushing the model downwards. After calculating the coefficients, the data was then mapped to an equation of

the form $C(Re) = C_{\infty} + Ae^{-\frac{Re}{Re_0}}$, which allowed it to reach a constant value as the Reynolds number increased. This is consistent with the assumption that bluff bodies eventually reach a consistent drag and lift coefficient (Ford et al, 2020). Once a constant coefficient was calculated, Equation 5 was applied with scaled human height and area to calculate the drag and lift force in a real life scenario (Center for Disease Control, 2023).

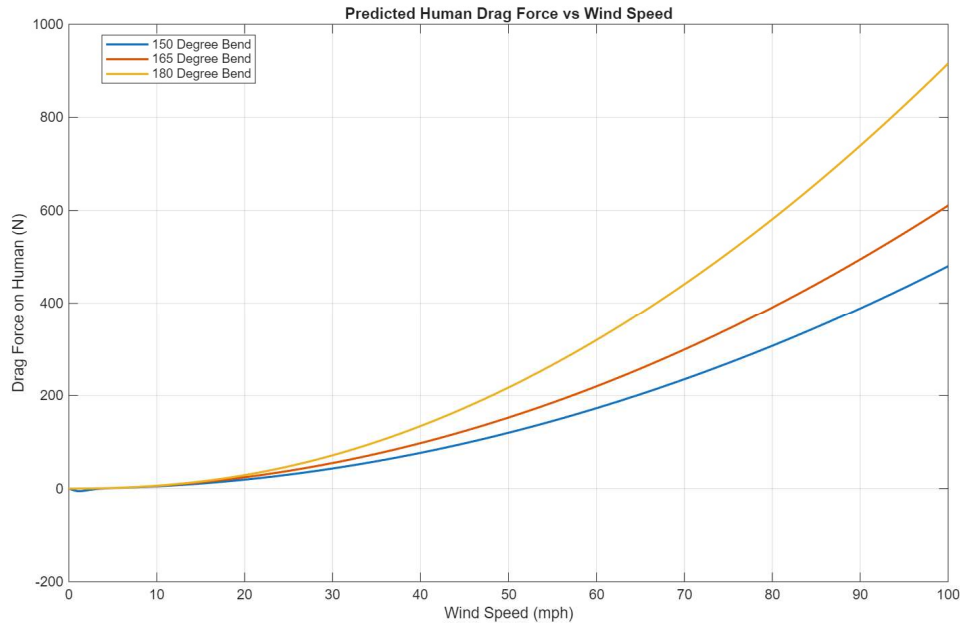


Fig. 4 Predicted full scale drag force versus wind speed for varying hip bend angles.

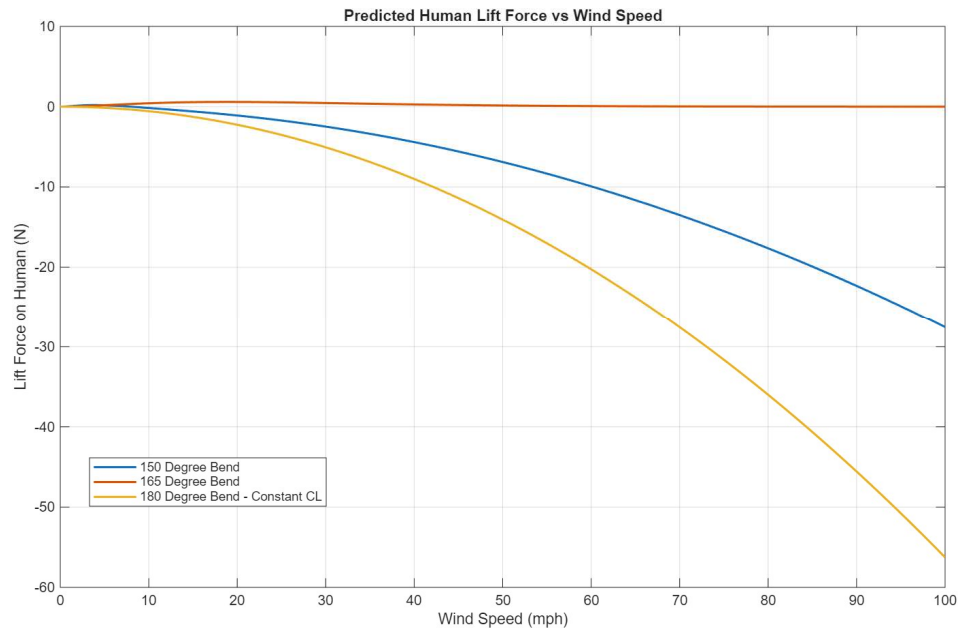


Fig. 5 Predicted full scale lift force versus wind speed for varying hip bend angles.

The results of the scaled-up calculations are consistent with the coefficient trends. While the drag coefficient of the 165° hip bend was positive, it was low and decreasing, leading to essentially no drag coefficient at scaled-up calculations. Finally, the tractive limit for each model was calculated by subtracting each lift force from the weight of an average human, then calculating the friction force and comparing to the drag force. A friction coefficient of 0.9 was assumed (Engineering Toolbox, 2004).

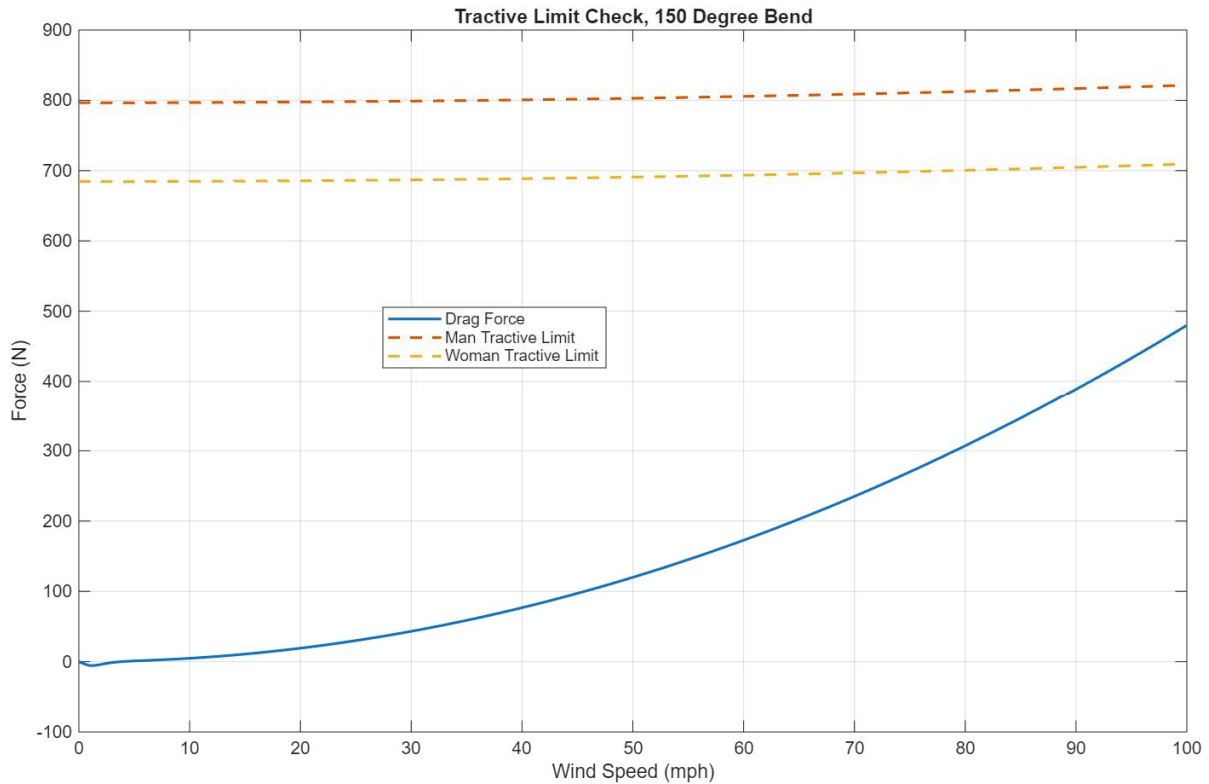


Fig. 6 Full scale drag force and tractive limits at 150° hip bend.

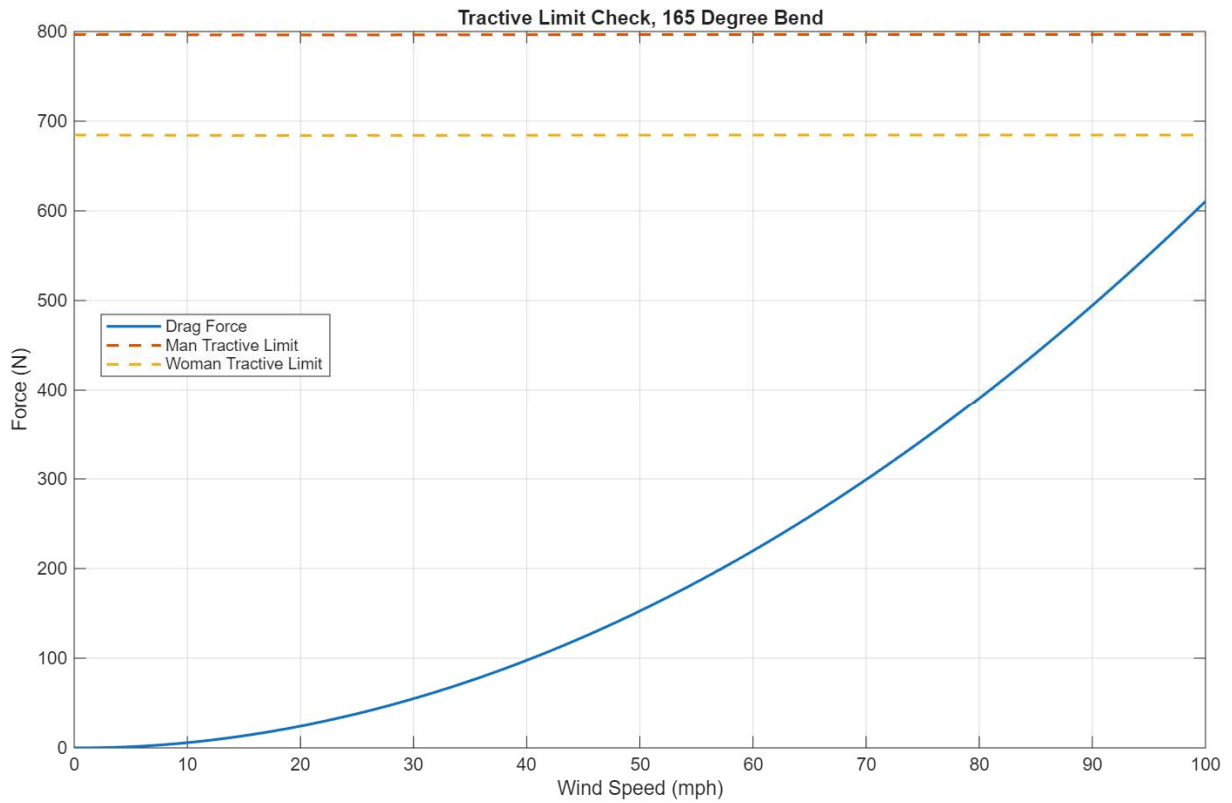


Fig. 7 Full scale drag force and tractive limits at 165° hip bend.

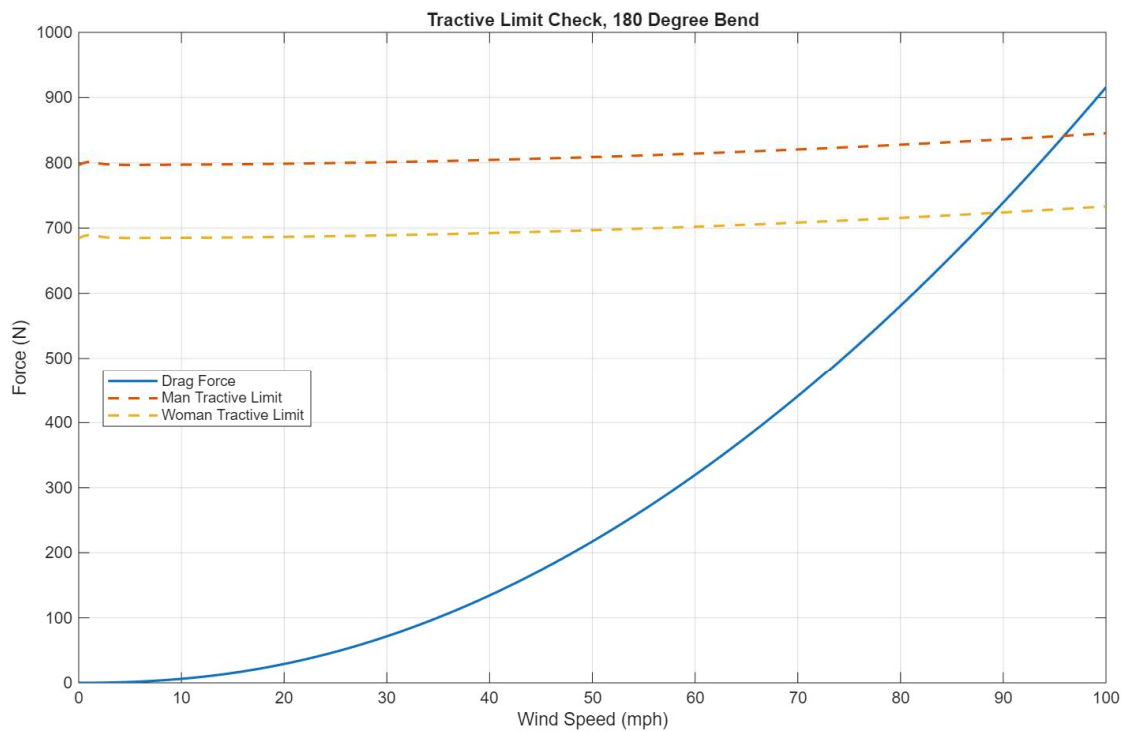


Fig. 8 Full scale drag force and tractive limits at 180° hip bend.

5 DISCUSSION

This experiment was able to meet the first objective of quantifying the lift and drag coefficients of the three small-scale human body models at $U \approx 22.67\text{--}34.01$ m/s and hip bend angle $\theta \in \{150^\circ, 165^\circ, 180^\circ\}$. As shown in Figure 2, the magnitude of the drag coefficient generally increased across various Reynolds numbers and was greatest for the 180° model with drag coefficients ranging from 0.85–0.97, least for the 150° model with drag coefficients ranging from 0.71–0.84, and the drag coefficient 165° model fell in between with drag coefficients ranging from 0.83–0.93. The differences in drag coefficients are attributed to the differences in geometry between the three models. Larger surface area exposed to flow increases skin-friction drag, and changes in geometry affects how boundary layers form. While it is expected that the models have similar surface areas exposed to airflow, a lower hip bend angle model may allow for flow separation to be delayed as it is more streamlined as compared to a hip bend angle closer to 180° degrees. In addition, tape was used to cover the bolt used to mount the model, which covered an increasingly significant area of the model as the hip bend angle decreased, meaning for this experiment, the model's torso area became less bluff as hip angle decreased, thus decreasing pressure drag. With that said, the 95% confidence intervals of the drag coefficient of the 165° model and the 180° model fell within each other across multiple points with an interference in drag coefficient of approximately 0.025. However, fluid mechanics principles suggest that differences were a result of the change in model rather than uncertainty of measurement.

The lift coefficient is quantified across various Reynolds numbers as shown in Fig. 3. As shown in the figure, lift coefficient was greatest for the 150° hip bend angle, in between for the 165° hip bend angle, and least for the 180° hip bend angle. The lift coefficients appeared to decrease across various Reynolds numbers for the 150° and 165° hip bend angle models with ranges of 0.05–0.11 and -0.03–0.12, respectively. For the 180° hip bend angle model, lift coefficient increased until reaching a Reynolds number of approximately $3.5 \cdot 10^5$ before decreasing again; lift coefficients ranged from -0.11–0.025. It was expected that the 180° model had the lowest lift coefficient across all Reynolds numbers as it has a 90° angle of attack and was most certainly above the critical angle of attack. As for the 150° and 165° models, lift coefficient generally decreased as Reynolds' number increased. This can be explained by Reynolds' number effects; higher Reynolds' number can change the boundary layer behavior, separation point, and pressure distribution over the models thus changing the lift coefficient.

The second objective to identify which hip bend angle produces the least lift and the least drag for each wind speed was also met. Fig. 4 and Fig. 5 show the predicted drag force vs wind speed and lift force. The order of drag force with respect to all three models is the same as the order of drag coefficients for all flow speeds. Fluid mechanics principles suggest that this should be the case as drag force is affected by drag coefficient as well as planform area, fluid density, and flow velocity. Since fluid density is assumed to stay the same and flow velocity is a controlled factor, an increase in planform area will result in drag force also increasing. Thus, for all wind speeds, the 180° model, which the greatest planform area and highest drag coefficient, had the highest predicted drag force, the 165° model had predicted drag force in between the 180° and 150° models, and the 150° model had the least predicted drag force. The predicted drag forces at 40 mph, where the wind generally impedes walking (National Weather Service, n.d.) for the 180° , 165° , and 150° models are ≈ 315 N, ≈ 212 N, and ≈ 186 N, respectively.

The third objective of predicting where the typical person experiences unsafe conditions either due to liftoff or the tractive limit being exceeded by drag force was also met. It was found that for the models tested, lift force decreased with increasing wind speed, thus liftoff was not considered

to be a safety issue. However, the tractive limit being exceeded was considered to pose safety concerns as drag force increased with wind speed. **Fig. 6** (150° model), **Fig. 7** (165° model), and **Fig. 8** (180° model), show the drag force of each model with the tractive limit overlaid for wind speeds of 0–100 mph. In **Fig. 6** and **Fig. 7**, it is shown that predicted drag force does not exceed the tractive limit for men or women for windspeeds less than 100 mph. **Fig. 8** shows that the drag force exceeds the tractive limit at ≈ 88 mph for women and at ≈ 96 mph for men.

Some limitations of the experiment include the mounting interference with airflow, measuring wind which has quasi-steady behavior, wind tunnel limitations, model limitations, low Reynolds number forcing incomplete similarity, and imperfect ground effect consideration.

To secure the model to the test stand, a threaded rod and nut were utilized. However, to reduce the effect the nut had on the flow characteristics, it was taped over. This change to the model affects flow characteristics in other ways; as hip angle was decreased, more tape was used to cover the model as the bolt was positioned such that it required more tape to cover. This change to the model may affect the skin friction drag at the front of the model due to change in material as well as flow separation given that the nut or tape weren't present. It is predicted that the addition of the nut to mount the model in addition to the tape decreased the drag force on the models due to the addition decreasing how bluff the models were and thus changing the point of flow separation. In addition, the threaded rod at the back affects the airflow at the back of the model. Without the rod, a larger turbulent wake with lower pressure would form thus increasing form drag and drag force. To address the mounting interference issue, it is recommended to design a custom mounting system such that the nut affects the airflow less, perhaps by making it possible for the nut to sit inside the model and such that the material the model is made of could be used to cover the nut. In addition, it is recommended to quantify the effects of threaded rod protruding out the back of the model and to adjust accordingly.

In reality, it is more common to experience instability in walking due to unpredictable gusts rather than constant wind speeds. The National Weather Service quantifies frequent gusts in the range of 39–55 mph as unsafe for walking. However, due to the sampling speed of dynamometer as well as the limitations of how fast frequency can safely be changed in the wind tunnel, testing dynamic wind conditions was not possible. To improve this experiment in the future, it would be important to obtain equipment such that dynamic wind conditions could be tested.

A limitation with the model as compared to a human is the presence of flexible and elastic clothing with significantly different surface characteristics than 3D-printed PLA. Differences in surface characteristics can significantly affect skin friction drag and clothing also increases the surface area that the flow reaches. It is predicted that the model experiences less drag force than a model with clothing would. To address this limitation, it is recommended that clothing for the model is fabricated to match typical human clothing such that the surface characteristics and increase in area are accounted for.

Since the model was much smaller than a human being, and the wind tunnel had a max speed much lower than the situation addressed, this experiment was forced to use incomplete similarity. However, we cannot be certain that the models act as perfect bluff bodies, meaning that our approximation of a constant drag coefficient may have been inaccurate. To address this limitation, future iterations of this project could use a wind tunnel with higher max wind speeds.

While the model was mounted close to the ground, the feet were placed off the ground to allow the dynamometer to read force. This affects flow characteristics as ground effects are not fully considered since the feet are not constrained to the ground. This may affect the lift force as air is

able to flow below the feet. To mitigate this limitation, it is recommended to simulate the model and determine an appropriate compensation factor to adjust the lift coefficient.

Another limitation with this experiment setup is that it does not account for varying wind directions. If the project were repeated, a suggestion would be to increase the model variety such that models can be mounted at various angles with respect to the air direction as opposed to directly facing the wind.

6 CONCLUSION

In conclusion, all objectives of the experiment were met including determining the drag and lift coefficients of three small-scale human body models at free stream velocities of approximately 22.67–34.01 m/s and hip bend angles of 150°, 165°, and 180° as shown in Figures 2 and 3, identifying which hip bend angle produces the least predicted lift and the least drag for each wind speed as shown in Figures 4, and 5, and identifying the minimum wind speed where safety concerns begin to formulate due to lift force exceeding weight or drag force exceeding tractive limit, as shown in Figures 6, 7, and 8. An important quantitative takeaway includes the predicted drag forces on a typical human subject at the at 40 mph, where the wind generally impedes walking (National Weather Service, n.d.); for the 180°, 165°, and 150° models the drag forces are ≈ 315 N, ≈ 212 N, and ≈ 186 N, respectively. Drag force is shown to be least for the 150° model, and most for the 180° model. Additionally, it is shown that the tractive limit is exceeded by the drag force at ≈ 88 mph for women and at ≈ 96 mph for men for the 180° model. Some important practical implications of this experiment is that for steady wind behavior, it would require very high speeds such that the drag force exceeds the tractive limit, and that lift force is not a safety concern, as it decreases with increasing wind speed.

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APPENDIX

APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

Item\Name	Luo, Henry	May, Janine	O-Lee, Sean	Subtotal
Project Idea	25%	50%	25%	100%
Proposal Preparation	33%	33%	33%	100%
Lab Test	33%	33%	33%	100%
Data Processing			100%	100%
Report Writing				
Introduction	100%			100%
Objective	100%			100%
Method	100%			100%
Results		10%	90%	100%
Discussion		100%		100%
Edit & Proofread	30%	35%	35%	100%
PPT Preparation	40%	30%	30%	100%

APPENDIX B: CAD DRAWINGS

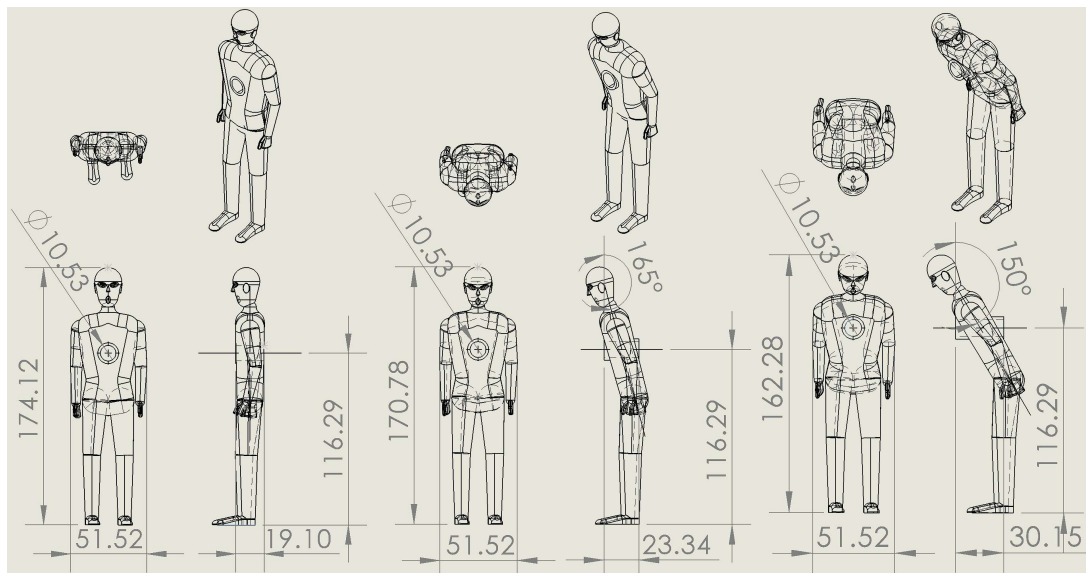


Fig. 9 Small-scale human test models. From left to right: model with 180° hip bend, model with 165° hip bend, and model with 150° hip bend. Models are scaled, bent, and aesthetically-altered from GrabCAD using SolidWorks (Neubern, 2025).

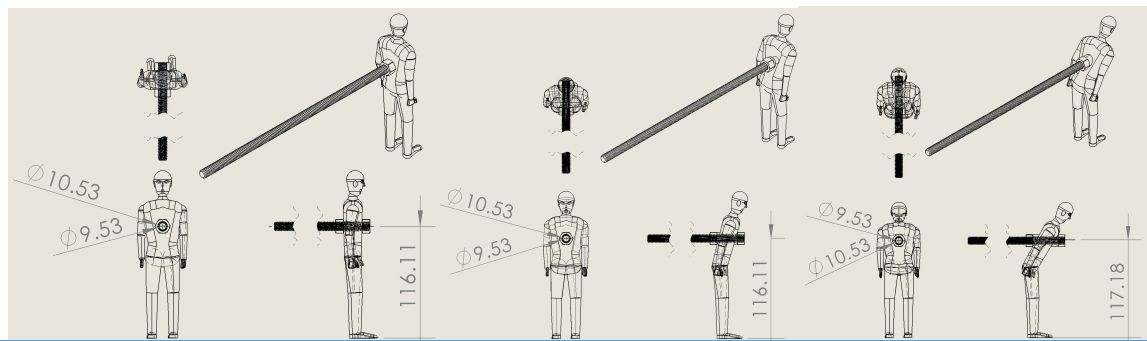


Fig. 10 Mounting of small-scale human test models. From left to right: model with 180° hip bend, model with 165° hip bend, and model with 150° hip bend. The floor (blue bar) is to be adjusted to provide as little ground clearance as possible without touching the model's feet when the model is shaken.


```

1  clear; clc;
2
3  dragscalingfactor = 12.52;
4  liftscalingfactor = 89.49;
5  dragscalingfactor2 = 12.47;
6  liftscalingfactor2 = 72.48;
7
8  toolUncertainty = 0.01;    % N
9
10 data150    = readmatrix('150deg.xlsx');
11 data165    = readmatrix('165deg.xlsx');
12 data180    = readmatrix('180deg.xlsx');
13 datastand  = readmatrix('teststand1data.xlsx');
14 datastand2 = readmatrix('teststand2data.xlsx');
15
16 area150 = 5.02682e-3;
17 area165 = 5.28867e-3;
18 area180 = 5.37630e-3;
19
20 n = 10;
21 t95 = 2.262;
22
23 vel150     = data150(:,2);
24 drag150     = data150(:,3);
25 dragstd150 = data150(:,4);
26 lift150     = data150(:,5);
27 liftstd150 = data150(:,6);
28
29 vel165     = data165(:,2);
30 drag165     = data165(:,3);
31 dragstd165 = data165(:,4);
32 lift165     = data165(:,5);
33 liftstd165 = data165(:,6);
34
35 vel180     = data180(:,2);
36 drag180     = data180(:,3);
37 dragstd180 = data180(:,4);
38 lift180     = data180(:,5);
39 liftstd180 = data180(:,6);
40
41 velstand    = datastand(:,2);
42 dragstand    = datastand(:,3);
43 dragstdstand = datastand(:,4);
44 liftstand    = datastand(:,5);
45 liftstdstand = datastand(:,6);
46
47 velstand2    = datastand2(:,2);
48 dragstand2    = datastand2(:,3);
49 dragstdstand2 = datastand2(:,4);
50 liftstand2    = datastand2(:,5);
51 liftstdstand2 = datastand2(:,6);
52
53 % Force calculations
54 forcedrag165 = zeros(length(drag165),1);
55 forcelif165  = zeros(length(lift165),1);
56 forcedrag180 = zeros(length(drag180),1);
57 forcelif180  = zeros(length(lift180),1);
58 forcedrag150 = zeros(length(drag150),1);
59 forcelif150  = zeros(length(lift150),1);
60
61 for i = 1:length(vel165)

```

```

62     forcedrag165(i) = dragscalingfactor * ((drag165(i) - drag165(1)) - (dragstand(i) -
dragstand(1)));
63     forcelif165(i) = liftscalingfactor * ((lift165(i) - lift165(1)) - (liftstand(i) -
liftstand(1)));
64 end
65
66 for i = 1:length(vel180)
67     forcedrag180(i) = dragscalingfactor * ((drag180(i) - drag180(1)) - (dragstand(i) -
dragstand(1)));
68     forcelif180(i) = liftscalingfactor * ((lift180(i) - lift180(1)) - (liftstand(i) -
liftstand(1)));
69 end
70
71 for i = 1:length(vel150)
72     forcedrag150(i) = dragscalingfactor2 * ((drag150(i) - drag150(1)) - (dragstand2(i) -
dragstand2(1)));
73     forcelif150(i) = liftscalingfactor2 * ((lift150(i) - lift150(1)) - (liftstand2(i) -
liftstand2(1)));
74 end
75
76 % 95% confidence interval in FORCE
77 dragCIforce150 = zeros(length(drag150),1);
78 liftCIforce150 = zeros(length(lift150),1);
79 dragCIforce165 = zeros(length(drag165),1);
80 liftCIforce165 = zeros(length(lift165),1);
81 dragCIforce180 = zeros(length(drag180),1);
82 liftCIforce180 = zeros(length(lift180),1);
83
84 for i = 1:length(vel150)
85     se_drag_150 = sqrt((dragstd150(i)^2 + dragstd150(1)^2 + dragstdstand2(i)^2 +
dragstdstand2(1)^2) / n);
86     se_lift_150 = sqrt((liftstd150(i)^2 + liftstd150(1)^2 + liftstdstand2(i)^2 +
liftstdstand2(1)^2) / n);
87     dragCIforce150(i) = sqrt((t95 * dragscalingfactor2 * se_drag_150)^2 + toolUncertainty^2);
88     liftCIforce150(i) = sqrt((t95 * liftscalingfactor2 * se_lift_150)^2 + toolUncertainty^2);
89 end
90
91 for i = 1:length(vel165)
92     se_drag_165 = sqrt((dragstd165(i)^2 + dragstd165(1)^2 + dragstdstand(i)^2 +
dragstdstand(1)^2) / n);
93     se_lift_165 = sqrt((liftstd165(i)^2 + liftstd165(1)^2 + liftstdstand(i)^2 +
liftstdstand(1)^2) / n);
94     dragCIforce165(i) = sqrt((t95 * dragscalingfactor * se_drag_165)^2 + toolUncertainty^2);
95     liftCIforce165(i) = sqrt((t95 * liftscalingfactor * se_lift_165)^2 + toolUncertainty^2);
96 end
97
98 for i = 1:length(vel180)
99     se_drag_180 = sqrt((dragstd180(i)^2 + dragstd180(1)^2 + dragstdstand(i)^2 +
dragstdstand(1)^2) / n);
100    se_lift_180 = sqrt((liftstd180(i)^2 + liftstd180(1)^2 + liftstdstand(i)^2 +
liftstdstand(1)^2) / n);
101    dragCIforce180(i) = sqrt((t95 * dragscalingfactor * se_drag_180)^2 + toolUncertainty^2);
102    liftCIforce180(i) = sqrt((t95 * liftscalingfactor * se_lift_180)^2 + toolUncertainty^2);
103 end
104
105 % Coefficients
106 cd150 = zeros(length(forcedrag150),1);
107 cl150 = zeros(length(forcelif150),1);
108 cd165 = zeros(length(forcedrag165),1);
109 cl165 = zeros(length(forcelif165),1);
110 cd180 = zeros(length(forcedrag180),1);

```

```

111 cl180 = zeros(length(forcelift180),1);
112
113 cdCI150 = zeros(length(forcedrag150),1);
114 clCI150 = zeros(length(forcelift150),1);
115 cdCI165 = zeros(length(forcedrag165),1);
116 clCI165 = zeros(length(forcelift165),1);
117 cdCI180 = zeros(length(forcedrag180),1);
118 clCI180 = zeros(length(forcelift180),1);
119
120 p1 = 1.184;
121 p2 = 1.176;
122
123 for i = 1:length(forcedrag150)
124     if vel150(i) == 0
125         cd150(i) = NaN;
126         cl150(i) = NaN;
127         cdCI150(i) = NaN;
128         clCI150(i) = NaN;
129     else
130         denom150 = p2 * vel150(i)^2 * area150;
131         cd150(i) = 2 * forcedrag150(i) / denom150;
132         cl150(i) = 2 * forcelift150(i) / denom150;
133         cdCI150(i) = 2 * dragCIforce150(i) / denom150;
134         clCI150(i) = 2 * liftCIforce150(i) / denom150;
135     end
136
137     if vel165(i) == 0
138         cd165(i) = NaN;
139         cl165(i) = NaN;
140         cdCI165(i) = NaN;
141         clCI165(i) = NaN;
142     else
143         denom165 = p1 * vel165(i)^2 * area165;
144         cd165(i) = 2 * forcedrag165(i) / denom165;
145         cl165(i) = 2 * forcelift165(i) / denom165;
146         cdCI165(i) = 2 * dragCIforce165(i) / denom165;
147         clCI165(i) = 2 * liftCIforce165(i) / denom165;
148     end
149
150     if vel180(i) == 0
151         cd180(i) = NaN;
152         cl180(i) = NaN;
153         cdCI180(i) = NaN;
154         clCI180(i) = NaN;
155     else
156         denom180 = p1 * vel180(i)^2 * area180;
157         cd180(i) = 2 * forcedrag180(i) / denom180;
158         cl180(i) = 2 * forcelift180(i) / denom180;
159         cdCI180(i) = 2 * dragCIforce180(i) / denom180;
160         clCI180(i) = 2 * liftCIforce180(i) / denom180;
161     end
162 end
163
164 L180 = 0.1743;
165 L165 = 0.17096;
166 L150 = 0.16245;
167
168 Re150 = vel150 * L150 / (1.55e-5);
169 Re165 = vel165 * L165 / (1.54e-5);
170 Re180 = vel180 * L180 / (1.54e-5);
171

```

```

172 % Drag coefficient plot with 95% CI error bars
173 figure;
174 errorbar(Re150, cd150, cdCI150, '-o', 'LineWidth', 1.2, 'CapSize', 6);
175 hold on;
176 errorbar(Re165, cd165, cdCI165, '-o', 'LineWidth', 1.2, 'CapSize', 6);
177 errorbar(Re180, cd180, cdCI180, '-o', 'LineWidth', 1.2, 'CapSize', 6);
178 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
179 xlabel('Reynolds Number');
180 ylabel('Drag Coefficient');
181 title('Drag Coefficient vs Reynolds Number for Varying Hip Bend Angles');
182 grid on;
183
184 % Lift coefficient plot with 95% CI error bars
185 figure;
186 errorbar(Re150, cl150, clCI150, '-o', 'LineWidth', 1.2, 'CapSize', 6);
187 hold on;
188 errorbar(Re165, cl165, clCI165, '-o', 'LineWidth', 1.2, 'CapSize', 6);
189 errorbar(Re180, cl180, clCI180, '-o', 'LineWidth', 1.2, 'CapSize', 6);
190 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
191 xlabel('Reynolds Number');
192 ylabel('Lift Coefficient');
193 title('Lift Coefficient vs Reynolds Number for Varying Hip Bend Angles');
194 grid on;
195
196
197 %% Human-scale wind force prediction
198
199 scale = 10;
200
201 Lhuman150 = scale * L150;
202 Lhuman165 = scale * L165;
203 Lhuman180 = scale * L180;
204
205 Ahuman150 = scale^2 * area150;
206 Ahuman165 = scale^2 * area165;
207 Ahuman180 = scale^2 * area180;
208
209 rhoHuman = 1.184;
210 nuHuman = 1.54e-5;
211
212 Vhuman_mph = linspace(0, 100, 200);
213 Vhuman = Vhuman_mph * 0.44704;
214
215 % Drag fits use original saturating fit
216 dragFit150 = fitSaturatingCoeff(Re150, cd150);
217 dragFit165 = fitSaturatingCoeff(Re165, cd165);
218 dragFit180 = fitSaturatingCoeff(Re180, cd180);
219
220 % Lift fits
221 % Only 165 degree bend uses positive large-Re fit
222 % 180 degree bend uses constant CL to smooth lift force
223 liftFit150 = fitSaturatingCoeff(Re150, cl150);
224 liftFit165 = fitPositiveSaturatingCoeff(Re165, cl165);
225 CL180_const = mean(cl180(isfinite(cl180)));
226
227 ReHuman150 = Vhuman * Lhuman150 / nuHuman;
228 ReHuman165 = Vhuman * Lhuman165 / nuHuman;
229 ReHuman180 = Vhuman * Lhuman180 / nuHuman;
230
231 CdHuman150 = evalSaturatingCoeff(dragFit150, ReHuman150);
232 CdHuman165 = evalSaturatingCoeff(dragFit165, ReHuman165);

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233 CdHuman180 = evalSaturatingCoeff(dragFit180, ReHuman180);
234
235 ClHuman150 = evalSaturatingCoeff(liftFit150, ReHuman150);
236 ClHuman165 = evalSaturatingCoeff(liftFit165, ReHuman165);
237 ClHuman180 = CL180_const * ones(size(Vhuman));
238
239 DragHuman150 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman150 .* CdHuman150;
240 DragHuman165 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman165 .* CdHuman165;
241 DragHuman180 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman180 .* CdHuman180;
242
243 LiftHuman150 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman150 .* ClHuman150;
244 LiftHuman165 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman165 .* ClHuman165;
245 LiftHuman180 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman180 .* ClHuman180;
246
247 figure;
248 plot(Vhuman_mph, DragHuman150, 'LineWidth', 1.5);
249 hold on;
250 plot(Vhuman_mph, DragHuman165, 'LineWidth', 1.5);
251 plot(Vhuman_mph, DragHuman180, 'LineWidth', 1.5);
252 xlabel('Wind Speed (mph)');
253 ylabel('Drag Force on Human (N)');
254 title('Predicted Human Drag Force vs Wind Speed');
255 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
256 grid on;
257
258 figure;
259 plot(Vhuman_mph, LiftHuman150, 'LineWidth', 1.5);
260 hold on;
261 plot(Vhuman_mph, LiftHuman165, 'LineWidth', 1.5);
262 plot(Vhuman_mph, LiftHuman180, 'LineWidth', 1.5);
263 xlabel('Wind Speed (mph)');
264 ylabel('Lift Force on Human (N)');
265 title('Predicted Human Lift Force vs Wind Speed');
266 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend - Constant CL', 'Location',
'best');
267 grid on;
268
269 figure;
270 plot(Vhuman_mph, ClHuman150, 'LineWidth', 1.5);
271 hold on;
272 plot(Vhuman_mph, ClHuman165, 'LineWidth', 1.5);
273 plot(Vhuman_mph, ClHuman180, 'LineWidth', 1.5);
274 xlabel('Wind Speed (mph)');
275 ylabel('Extrapolated Lift Coefficient');
276 title('Human-Scale Lift Coefficient vs Wind Speed');
277 legend('150 Degree Bend', '165 Degree Bend - Positive Fit', '180 Degree Bend - Constant CL',
'Location', 'best');
278 grid on;
279
280 fprintf('\nDrag coefficient equations: C_D = C_inf + A*exp(-Re/Re0)\n');
281 fprintf('150 deg: C_D = %.4f + %.4f*exp(-Re/%.4e)\n', dragFit150(1), dragFit150(2),
dragFit150(3));
282 fprintf('165 deg: C_D = %.4f + %.4f*exp(-Re/%.4e)\n', dragFit165(1), dragFit165(2),
dragFit165(3));
283 fprintf('180 deg: C_D = %.4f + %.4f*exp(-Re/%.4e)\n', dragFit180(1), dragFit180(2),
dragFit180(3));
284
285 fprintf('\nLift coefficient equations:\n');
286 fprintf('150 deg: C_L = %.4f + %.4f*exp(-Re/%.4e)\n', liftFit150(1), liftFit150(2),
liftFit150(3));

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287 fprintf('165 deg: C_L = %.4f + %.4f*exp(-Re/%.4e)  positive large-Re fit\n', liftFit165(1),
liftFit165(2), liftFit165(3));
288 fprintf('180 deg: C_L = %.4f  constant smooth fit\n', CL180_const);
289
290 fprintf('\nHuman-scale CL ranges:\n');
291 fprintf('150 deg CL: min = %.4f, max = %.4f\n', min(CLHuman150), max(CLHuman150));
292 fprintf('165 deg CL: min = %.4f, max = %.4f\n', min(CLHuman165), max(CLHuman165));
293 fprintf('180 deg CL: min = %.4f, max = %.4f\n', min(CLHuman180), max(CLHuman180));
294
295
296 %% Unsafe wind speed check using tractive limit
297
298 mu = 0.9;
299
300 weightMan_lb = 199;
301 weightWoman_lb = 171;
302
303 lb_to_N = 4.44822;
304
305 Wman = weightMan_lb * lb_to_N;
306 Wwoman = weightWoman_lb * lb_to_N;
307
308 tractiveMan150 = mu * (Wman - LiftHuman150);
309 tractiveMan165 = mu * (Wman - LiftHuman165);
310 tractiveMan180 = mu * (Wman - LiftHuman180);
311
312 tractiveWoman150 = mu * (Wwoman - LiftHuman150);
313 tractiveWoman165 = mu * (Wwoman - LiftHuman165);
314 tractiveWoman180 = mu * (Wwoman - LiftHuman180);
315
316 unsafeMan150 = DragHuman150 > tractiveMan150;
317 unsafeMan165 = DragHuman165 > tractiveMan165;
318 unsafeMan180 = DragHuman180 > tractiveMan180;
319
320 unsafeWoman150 = DragHuman150 > tractiveWoman150;
321 unsafeWoman165 = DragHuman165 > tractiveWoman165;
322 unsafeWoman180 = DragHuman180 > tractiveWoman180;
323
324 unsafeWindMan150 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan150);
325 unsafeWindMan165 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan165);
326 unsafeWindMan180 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan180);
327
328 unsafeWindWoman150 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman150);
329 unsafeWindWoman165 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman165);
330 unsafeWindWoman180 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman180);
331
332 fprintf('\nUnsafe wind speeds using tractive limit:\n');
333 fprintf('Unsafe when Drag Force > mu*(Weight - Lift Force)\n');
334 fprintf('mu = %.2f\n\n', mu);
335
336 fprintf('Man, %.0f lb:\n', weightMan_lb);
337 fprintf('150 deg: %.2f mph\n', unsafeWindMan150);
338 fprintf('165 deg: %.2f mph\n', unsafeWindMan165);
339 fprintf('180 deg: %.2f mph\n\n', unsafeWindMan180);
340
341 fprintf('Woman, %.0f lb:\n', weightWoman_lb);
342 fprintf('150 deg: %.2f mph\n', unsafeWindWoman150);
343 fprintf('165 deg: %.2f mph\n', unsafeWindWoman165);
344 fprintf('180 deg: %.2f mph\n', unsafeWindWoman180);
345
346 figure;

```

```

347 plot(Vhuman_mph, DragHuman150, 'LineWidth', 1.5);
348 hold on;
349 plot(Vhuman_mph, tractiveMan150, '--', 'LineWidth', 1.5);
350 plot(Vhuman_mph, tractiveWoman150, '--', 'LineWidth', 1.5);
351 xlabel('Wind Speed (mph)');
352 ylabel('Force (N)');
353 title('Tractive Limit Check, 150 Degree Bend');
354 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
355 grid on;
356
357 figure;
358 plot(Vhuman_mph, DragHuman165, 'LineWidth', 1.5);
359 hold on;
360 plot(Vhuman_mph, tractiveMan165, '--', 'LineWidth', 1.5);
361 plot(Vhuman_mph, tractiveWoman165, '--', 'LineWidth', 1.5);
362 xlabel('Wind Speed (mph)');
363 ylabel('Force (N)');
364 title('Tractive Limit Check, 165 Degree Bend');
365 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
366 grid on;
367
368 figure;
369 plot(Vhuman_mph, DragHuman180, 'LineWidth', 1.5);
370 hold on;
371 plot(Vhuman_mph, tractiveMan180, '--', 'LineWidth', 1.5);
372 plot(Vhuman_mph, tractiveWoman180, '--', 'LineWidth', 1.5);
373 xlabel('Wind Speed (mph)');
374 ylabel('Force (N)');
375 title('Tractive Limit Check, 180 Degree Bend');
376 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
377 grid on;
378
379
380 %% Local functions
381
382 function params = fitSaturatingCoeff(Re, C)
383     valid = isfinite(Re) & isfinite(C) & Re > 0;
384     Re = Re(valid);
385     C = C(valid);
386
387     Cinf0 = mean(C(end-2:end));
388     A0 = C(1) - Cinf0;
389     Re00 = mean(Re);
390
391     p0 = [Cinf0, A0, Re00];
392
393     obj = @(p) sum((C - (p(1) + p(2)*exp(-Re./abs(p(3)))).^2);
394
395     params = fminsearch(obj, p0);
396     params(3) = abs(params(3));
397 end
398
399 function params = fitPositiveSaturatingCoeff(Re, C)
400     valid = isfinite(Re) & isfinite(C) & Re > 0 & C > 0;
401     Re = Re(valid);
402     C = C(valid);
403
404     Cinf0 = max(mean(C(end-2:end)), 0.001);
405     A0 = C(1) - Cinf0;
406     Re00 = mean(Re);
407

```

```

408     p0 = [log(Cinf0), A0, log(Re00)];
409
410     obj = @(p) sum((C - (exp(p(1)) + p(2)*exp(-Re./exp(p(3)))).^2);
411
412     p = fminsearch(obj, p0);
413
414     Cinf = exp(p(1));
415     A     = p(2);
416     Re0   = exp(p(3));
417
418     params = [Cinf, A, Re0];
419 end
420
421 function C = evalSaturatingCoeff(params, Re)
422     Cinf = params(1);
423     A     = params(2);
424     Re0   = params(3);
425
426     C = Cinf + A .* exp(-Re ./ Re0);
427 end
428
429 function unsafeSpeed = findFirstUnsafeSpeed(V, unsafeArray)
430     idx = find(unsafeArray, 1, 'first');
431
432     if isempty(idx)
433         unsafeSpeed = NaN;
434     else
435         unsafeSpeed = V(idx);
436     end
437 end

```